Decision-Aware Model-Family Selection for Time Series Foundation Models

Research proposal: cheap selection under asymmetric quantile decisions

Core research question. Given a new time series and a known asymmetric downstream cost, can cheap pre-forecast properties predict whether a classical model family or a time-series foundation model (TSFM) family will minimize downstream decision cost, without running expensive validation across both families? When the cheap descriptor is insufficient, can the selector detect that and defer to full validation?

1 Relevant Past Papers

Wang et al., "Spectral Predictability as a Fast Reliability Indicator for Time Series Forecasting Model Selection" (2025). Proposes spectral predictability Ω, a fast power-spectrum-derived statistic, and shows across controlled datasets and 28 GIFT-Eval datasets that zero-shot TSFMs gain relative advantage in high-Ω regimes, motivating Ω as a first-pass family-selection rule [1]. **Gap:** the rule is evaluated using forecast-error metrics such as sMAPE and MSE, not downstream asymmetric decision cost. The authors state explicitly that "future work should examine probabilistic scores (CRPS, interval coverage) for additional validation." It therefore does not tell us whether the same family boundary remains valid when under- and over-prediction have unequal operational consequences. **Constraints we inherit:** the paper recommends Ω only for series longer than 1000 timesteps, names non-stationarity and exogenous-driven dynamics as conditions limiting its predictive power, and reports its own failure case at $\Omega = 0.66$ where statistical models beat zero-shot by nearly 50%.

Aksu et al., "GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation" (2024). Provides a large, leakage-aware benchmark spanning diverse domains, frequencies, statistical models, deep models, and TSFMs [2]. **Gap:** evaluates forecasting quality, not whether a cheap statistic can choose the decision-optimal family for a downstream cost function. It gives us the evaluation substrate, not the decision-aware selector. **Operational note:** the GIFT-Eval repository publishes per-model result files for over 100 models across 97 dataset configurations, which we exploit in K2 below.

Wang et al., "Time Series Forecastability Measures" (2025). Studies spectral predictability and the largest Lyapunov exponent as pre-model indicators of intrinsic forecastability and shows correlation with realized forecasting performance [3]. **Gap:** characterizes how forecastable a series is, not which model family should be deployed, and does not condition the answer on asymmetric decision costs.

Mandi et al., "Decision-Focused Learning: Through the Lens of Learning to Rank" (ICML 2022). Directly optimizes prediction models for downstream decision quality rather than prediction error, formalizing the broader point that better forecast metrics need not imply better decisions [4]. **Gap:** DFL changes or retrains the predictor and often interacts with the downstream optimizer during training. Our question is different: given existing forecasting families and a new dataset, can we cheaply select the right family before running exhaustive model validation?

"Why Model Selection Fails in Time Series Forecasting" (2026). Introduces a descriptor-based framework using trend strength, seasonality, noise level, and temporal dependence, and finds that rule-based selection "achieves low accuracy, with correct model identification occurring in only a small fraction of cases," with substantial ranking instability across datasets and horizons [5]. **Why this matters more than a related-work row:** this is the

strongest published challenge to the premise that cheap descriptors can guide model selection at all, and it must be positioned against directly rather than omitted. **Our distinction, verified against the full text:** the paper contains no treatment of decision cost, asymmetric loss, or cost sensitivity, and does not evaluate spectral predictability. It also attempts a harder task, naming the single best model among many candidates, whereas we choose between two families where a wrong choice is bounded by Δ. Their finding motivates K2 and K3 rather than invalidating the project, and it supplies the pivot described in Section 9.

Positioning. Existing work establishes four pieces separately: spectral structure can cheaply stratify forecasting performance; TSFM evaluation can be done at scale; decision quality can diverge from prediction quality; and descriptor-based selection is unreliable when evaluated on forecast error. The missing connection is a cheap, transferable, decision-aware model-family selector whose failure modes are explicitly characterized.

2 Motivation

The practical problem is model selection under deployment cost. A practitioner with a new time series can either validate many forecasting models, which is expensive, or use a cheap heuristic such as Ω, which was calibrated against forecasting error. The heuristic becomes questionable when the downstream loss is asymmetric: a forecast error that is harmless under symmetric loss may be very costly when under-provision and over-provision have different penalties.

For the asymmetric linear decision

$$C(q, y) = c_u(y - q)_+ + c_o(q - y)_+, \qquad \tau = \frac{c_u}{c_u + c_o},$$

the optimal action is the τ-quantile. Moreover,

$$C(q, y) = (c_u + c_o)\, \rho_\tau(y - q),$$

where ρ_τ is pinball loss. Minimizing downstream decision cost is therefore exactly equivalent, up to scale, to minimizing pinball loss at the operational quantile. This gives a precise expensive target: run all candidates, validate them under pinball-τ, and pick the best family.

Relation to generic probabilistic scoring. The same identity places CRPS and pinball-τ in an exact relationship:

$$\mathrm{CRPS}(F, y) \;=\; 2 \int_0^1 \rho_\tau\big(y - F^{-1}(\tau)\big)\; d\tau.$$

CRPS is therefore the uniform average of decision cost over *all* operational quantiles, while pinball-τ is the single slice that a given deployment actually faces. This makes the CRPS comparator in Section 5.1 a precise question rather than an arbitrary one: if the operational quantile carries no information, selection under CRPS and selection under pinball-τ agree, and the cost model can be ignored. If it does carry information, the two diverge, and the divergence should grow as τ moves away from 0.5. The Ω paper names probabilistic scoring, specifically CRPS and interval coverage, as future work; we run that comparison and additionally ask whether the τ-specific slice behaves differently from the integral.

Our aim is to approximate that expensive choice using information available before running the candidate models. Define

$$J_m(x, \tau) = \mathbb{E}[L_\tau(a_m(x), Y)], \qquad \Delta(x, \tau) = J_T(x, \tau) - J_C(x, \tau).$$

If $\Delta < 0$ the TSFM family is decision-optimal; otherwise the classical family is. The paper is about predicting the sign and practical magnitude of Δ cheaply.

Why this should work at all. The published Ω result suggests model-family advantage is structured rather than arbitrary. **Why the original rule may fail.** Ω summarizes spectral

and second-order temporal structure, whereas asymmetric quantile decisions can depend on distributional properties not represented by second-order statistics. We do not assume a specific "center versus tails" law. We test whether the boundary exists, whether it moves with τ, and exactly where Ω becomes insufficient.

2.1 A concrete instance of family-choice instability

Two oral papers at the same 2026 AI4TS workshop, presented on the same day by overlapping author teams from the same institute, reach opposite conclusions on Singapore electricity. One finds that Chronos, Lag-Llama, and Timer-XL beat statistical and deep-learning models on *price* by up to 37.4% in MAPE. The other finds that a RevIN-LSTM baseline beats foundation models in many cases on *demand*. Neither cites the other, and neither computes any structural descriptor that would explain the reversal.

This is the phenomenon our question addresses, already present in the literature and unexplained. We treat it as a named case study rather than a load-bearing result, because we may not obtain their exact preprocessing, and the paper must stand without it. If (Ω, τ) separates the two cases, the finding is quotable: a published contradiction resolved by a statistic that takes seconds to compute.

3 Key Ideas, Contributions, Novelty

1. **Decision-aware family boundary.** Replace a forecasting-error-based selection question with the exact downstream quantity $\Delta(x, \tau)$, and measure whether a stable family-choice boundary exists in the (Ω, τ) plane.
2. **Structural gate before fitting a correction.** A regression will always return a boundary even when the data do not support one. We first test monotone and order structure, and how much oracle decision value a single-Ω threshold can capture. If the boundary is not meaningful, we do not force a corrected threshold narrative.
3. **Controlled sufficiency test.** Construct matched-second-order synthetic processes with the same spectrum and hence matched Ω, but different innovation geometry. If they require different model families at the same (Ω, τ), then no deterministic Ω-only selector can distinguish them. This is a descriptor-insufficiency result, not merely a failed fitted model.
4. **Decision-aligned versus generic probabilistic validation.** We test whether selection under pinball-τ differs from selection under CRPS. A reviewer will ask why a well-calibrated predictive distribution is not sufficient on its own; this comparator answers that question directly and is computed at negligible cost from cached forecasts.
5. **Strict transfer and deployment value.** Freeze the selector on development datasets and test on entirely unseen M5 and PJM operational domains. Evaluate decision regret and compute saved, not only classification accuracy. A final abstention policy defers ambiguous cases to full pinball-τ validation.

The intended scientific contribution is not "a slightly better threshold." It is a characterization of when cheap pre-forecast information is sufficient for decision-aware model-family selection, when it is not, and what minimum additional information would repair the failure.

3.1 Novelty claim

That forecast quality and decision value can diverge is established [4] and we cite it as such. Three things are not established, and they are what this proposal claims.

Decision cost has not been used to select between frozen, zero-shot forecasters. Decision-focused learning changes or retrains the predictor and couples it to the downstream

optimizer during training [4]. Every published selection rule for pre-trained forecasters, including the Ω rule we extend, optimizes forecast error [1, 5]. Selecting among frozen candidates using the downstream cost, with no training at any stage, is a different operation and has not been done.

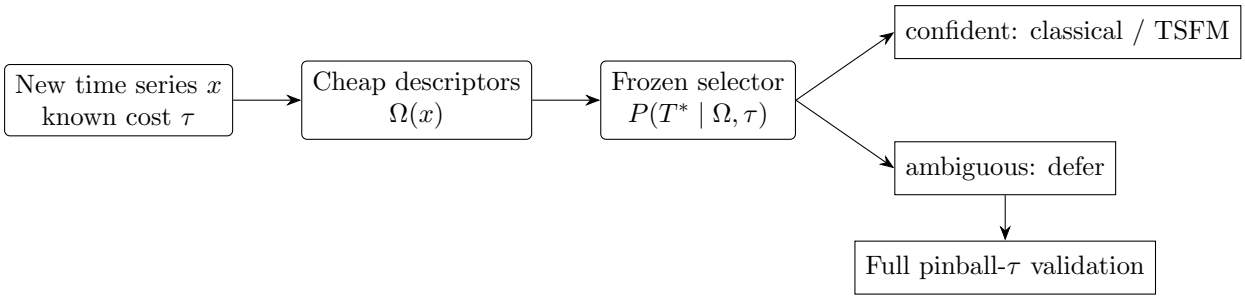
Whether the family boundary moves with the operational quantile is unmeasured. The published rule reports a fixed threshold calibrated at symmetric loss. Whether that threshold is a property of the data or of the cost structure imposed on it is an open empirical question, and $\Omega^*(\tau)$ is the object that answers it.

No prior work establishes a limit on what any Ω-only selector can achieve. The matched-Ω construction produces an insufficiency result about the descriptor rather than a negative result about one fitted model. That claim type is stronger and is not available from benchmark comparison alone.

Why this is not obvious in advance. That error-based and decision-based selection *can* disagree follows from the arithmetic in Section 2. Whether they *do* disagree, for these two families, on real data, at operationally realistic τ, depends on the empirical error geometry of the candidate models and cannot be derived. Both outcomes are informative: a moving boundary means a published rule is miscalibrated in a characterizable region, and a stable one is a robustness result nobody has demonstrated.

4 Methods

4.1 Pipeline



The expensive reference procedure is fixed: evaluate both candidate families using historical pinball-τ validation and choose the lower-cost family. The proposed selector sees only cheap series descriptors and τ at deployment time.

4.2 Required experiments

E0, premise and kill gates.

- **K1:** verify the published definition of Ω is determined by the second-order and spectral information assumed by the matched-Ω construction.
- **K2a, replicate on the original metric.** Reproduce the published Ω relationship on the metrics the source paper reports, namely sMAPE and MSE, using the per-model results GIFT-Eval already publishes. No forecasting model is run and no metric is changed. Given [5], replication of the premise is a precondition rather than a formality.
- **K2b, bridge to the decision metric.** Only after K2a passes, check whether the same relationship survives the move to pinball at $\tau = 0.5$. Separating these two steps matters: pinball at $\tau = 0.5$ is proportional to mean absolute error and is therefore not one of the metrics the original result was established on. Collapsing them would leave a failure ambiguous between a false premise, a broken pipeline, and a metric change. A K2b failure is itself informative, since it would show that metric choice alone moves the boundary.

- **Cost of these gates.** Both K2 steps are nearly free. The GIFT-Eval repository publishes per-model result files for over 100 models across 97 dataset configurations, so K2a reduces to computing Ω on the training splits and joining it to results that already exist.
- **K3:** for each pre-registered τ, compare monotone or isotonic and unconstrained estimates of $P(T^* \mid \Omega, \tau)$, count sign reversals in Δ, and measure the oracle value captured by the best scalar threshold. If a scalar boundary captures little value, stop fitting $\Omega^*(\tau)$ and reframe around insufficiency.

E1, decision-boundary sweep. Sweep τ on both sides of 0.5: $\tau \in \{0.01, 0.05, 0.09, 0.25, 0.40, 0.50, 0.60,$ The sweep costs no additional forecast compute, since pinball-τ is evaluated post hoc on cached quantile grids. Including the overage-dominant side is not merely a scope statement: the candidate boundary below is written in $|\tau - 0.5|$ and therefore *assumes* symmetry about 0.5. Sweeping both sides tests that assumption instead of asserting it. If the boundary proves asymmetric, the functional form is replaced by the smallest pre-registered alternative that admits different slopes on each side. If K3 passes, fit the smallest boundary model consistent with the data, beginning with

$$\Omega^*(\tau) = \Omega^*(0.5) + \beta\,|\tau - 0.5|.$$

The linear form is a minimal falsifiable model, not a law. If it clearly misfits, use the smallest pre-registered monotone alternative. Primary output: the (Ω, τ) decision map with empirical family advantage, the fitted boundary, and bootstrap uncertainty.

E2, frozen unseen-domain transfer. Fit the selector on development datasets only, freeze all coefficients and within-family rules, and evaluate on datasets not used for fitting. Primary metric $R_{\text{sel}} = J_{\hat{m}} - \min_m J_m$. Report normalized decision regret, gap to full pinball-τ validation, worst-decile per-series regret, and measured selector cost.

E3, matched-Ω sufficiency test. Use an analytically controlled process such as $x_t = \phi x_{t-1} + \epsilon_t$ with equal innovation variance and matched autocovariance and spectrum, while changing innovation skew or tail weight. Numerically verify the Ω match before evaluation. Test whether the same observed (Ω, τ) can correspond to different signs of Δ. This is the main mechanism experiment.

E4, selective validation. Convert selector confidence into a three-way decision: choose classical, choose TSFM, or defer to full pinball-τ validation. Plot fraction of expensive validations avoided against excess decision regret. This is the deployment result.

E5, one minimal repair. Stretch goal, out of scope for the current submission cycle. If E3 exposes a systematic failure tied to distributional information absent from Ω, add exactly one pre-registered cheap descriptor $A(x)$, defaulting to a robust upper and lower quantile-spread asymmetry statistic. Compare only $D_{\Omega}(\Omega, \tau)$ against $D_{\Omega+A}(\Omega, A, \tau)$, with no feature search. The repair passes only if it fixes the controlled failure and reduces unseen-domain regret. **We deliberately do not commit to running E5 for this deadline.** It is conditional on E3, it creates a branch late in the schedule, and the insufficiency result from E3 is publishable on its own. If E3 fires and time permits, E5 is the first thing added back.

5 Experimental Setup

5.1 Baselines and control groups

Only comparators that close a necessary logical question remain in the main experiment.

Comparator	Purpose
Always classical	Cheap fixed-family deployment floor
Always TSFM	Expensive fixed-family deployment floor
Published static Ω rule	Direct prior method being tested and corrected
Pinball-τ validation	Exact expensive decision-aligned selector; approximation target
CRPS validation	Generic probabilistic selector; tests whether decision alignment adds information
D_Ω	Proposed cheap decision-aware selector
Hindsight family oracle	Ceiling and regret-normalization reference

On CRPS. We retain CRPS in the main table rather than the appendix. The most likely reviewer objection to this paper is that a well-calibrated predictive distribution should already suffice, so that family choice is irrelevant once the τ-quantile is extracted correctly. CRPS-based selection is the direct test of that objection, it is computed at negligible cost from cached forecasts, and it costs one table row. Either decision-aligned validation beats generic probabilistic validation, which is the result we expect, or it does not, in which case our finding is that practitioners can use generic probabilistic validation and skip the cost model entirely. Both outcomes are worth reporting; failing to determine which is not.

Not in the main table: random selector, MSE selector, ensembles, conformalized candidates, rolling Ω, CVaR and repeated-stochastic reliability studies, or large model-count sweeps.

5.2 Models

The claim is family-level, not per-model. **The within-family rule is stated explicitly and frozen:** within each family, select the single candidate with the lowest mean pinball-τ on the development datasets only, at $\tau = 0.5$, and freeze that choice before any held-out evaluation. Development and held-out sets are disjoint, so no held-out information enters the choice. The rule is applied identically to both families, so neither is advantaged by a more permissive procedure. As a robustness check we repeat the headline comparison with a fixed representative per family declared before any evaluation, to confirm the result is not an artifact of within-family selection. *[To confirm: The team should confirm this rule before the roster is frozen. The alternative is a fixed representative per family as the primary rule, which is simpler to defend but more vulnerable to the objection that a weak family member was chosen.]*

TSFM family: zero-shot probabilistic TSFMs compatible with the chosen benchmark pipeline, with a provisional shortlist of Chronos and Chronos-Bolt, Moirai, and TimesFM where probabilistic quantiles can be obtained consistently.

Classical and lightweight family: probabilistic or distribution-wrapped classical forecasters such as Seasonal Naive, AutoETS, and AutoARIMA, plus one lightweight learned baseline if required to match the original Ω replication taxonomy.

[To confirm: The exact roster should be frozen after K2 so it stays compatible with the published Ω result and does not become a model-count sweep. The proposal should not claim specific models until probabilistic-output compatibility and compute are verified.]

5.3 Confound control and ablations

- Generate full predictive distributions once and cache them. Every selector is evaluated on the same forecast realizations, and the τ sweep is post hoc.
- Use the same horizons and temporal splits for all model families on a dataset.
- Fix the within-family rule before held-out evaluation so D_Ω is not secretly doing per-model tuning.
- Pre-register τ values before held-out tests.

- Respect the source paper's stated constraint that Ω is unreliable below 1000 timesteps. Datasets whose usable history falls below that threshold are excluded and the exclusion is reported.
- The key ablation is descriptor sufficiency: matched-Ω synthetic pairs isolate higher-order distributional structure while holding the spectral descriptor fixed.

5.4 Statistics and visualizations

Use moving and block bootstrap confidence intervals at the series and domain level. Use forecast-comparison tests only for the small set of headline comparisons, with multiple-comparison correction applied there rather than across every exploratory appendix number.

Main visualizations: the (Ω, τ) plane with empirical family decision advantage, fitted boundary, uncertainty, held-out failures, and matched-Ω counterexamples; and the selective-validation frontier, plotting fraction of full validations avoided against excess decision regret.

6 Datasets and Evaluation

Development datasets. A pre-registered diverse subset of 8 to 12 GIFT-Eval and Monash collections spanning multiple domains, frequencies, and statistical structures. Candidate collections include M4, Electricity, Solar, ETT, Jena Weather, Hospital and US Births, Hierarchical Sales, Loop Seattle and SZ-Taxi, and Bitbrains.

Held out entirely. M5 retail demand, where the downstream task is a newsvendor-style inventory decision under asymmetric overage and underage costs; and PJM electricity, where the downstream task is reserve procurement under asymmetric shortage and excess reserve penalties. No within-benchmark split such as M5 grocery to M5 household counts as cross-domain generalization.

[To confirm: The exact PJM reserve-procurement cost construction and public penalty ratios must be frozen before the main run. If a defensible public penalty mapping is unavailable, retain PJM as a quantile-cost stress test but do not claim monetary realism.]

Evaluation metrics. Primary: normalized decision regret $J_{\hat{m}} - \min_m J_m$. Also: gap to full pinball-τ validation on unseen datasets; fraction of hindsight-oracle selection value recovered; per-series regret distribution and worst-decile regret; monetary and operational cost for M5 and PJM where the cost model is defensible; selector compute measured as model evaluations avoided, wall-clock, and GPU-seconds. Family-selection accuracy is a secondary diagnostic only.

7 Benchmarks and Evaluation Sets

The benchmark ladder is deliberately minimal: always classical, always TSFM, published static Ω, CRPS validation, full pinball-τ validation, D_Ω, and the hindsight family oracle.

Two safe regions are reported separately. The **dominance region**, where one family wins nearly always regardless of Ω, so no selector is needed. And the **selector-reliable region**, where expected regret of the cheap selector is below a pre-registered tolerance, so full validation can be skipped. Their cross-domain variability is measured rather than assumed.

8 Ideal Results

The strongest outcome is a coherent evidence chain, not a higher average score.

1. **Structure.** The published symmetric-cost relation replicates, and a stable family-choice boundary exists in the (Ω, τ) plane.
2. **Decision shift.** The static Ω rule loses decision value as the operational quantile changes, while a minimal τ-aware correction captures the shift.

3. **Alignment matters.** Decision-aligned validation selects differently from generic probabilistic validation, establishing that the cost model carries information CRPS does not.
4. **Transfer.** A selector frozen on development datasets recovers most of the decision value of full pinball-τ validation on entirely unseen M5 and PJM data.
5. **Mechanism.** Matched-second-order processes show that identical (Ω, τ) can map to different decision-optimal families when higher-order distributional geometry differs.
6. **Deployment.** The selector can abstain on ambiguous series, avoiding a large fraction of expensive validation while adding little excess decision regret.

Hypothesis. Cheap spectral structure contains useful, transferable information about model-family choice, but it is not universally sufficient once downstream costs emphasize asymmetric quantiles. A cost-aware selector using τ can recover much of the value of exhaustive decision-aligned validation; where Ω is insufficient, the missing information should be identifiable.

[To confirm: Exact numerical success thresholds, for example the minimum fraction of oracle value recovered or validation runs avoided, should be pre-registered after the five-day feasibility sprint, using measured noise and compute rather than arbitrary targets.]

9 Potential Limitations and Risks

- **Premise risk.** [5] finds descriptor-based selection correct in only a small fraction of cases, and the Ω paper itself reports a failure case at $\Omega = 0.66$. If either generalizes, the foundation is unsound. K2 tests this in week one at near-zero cost. If K2 fails, the project reframes: error-based rule selection is known to be unreliable, and we ask whether decision-aligned selection is more reliable. That is arguably the stronger paper, and the pivot costs nothing because K2 runs before any forecast compute.
- Ω **misreads integrated processes.** A random walk scores $\Omega \approx 0.74$ in our implementation, above the region where TSFMs are expected to win, despite being the standard example of an unpredictable series, because the statistic rewards low-frequency energy and a drifting series is smooth. Whether Ω is computed on levels or first differences must be decided and justified on day one of the sprint, and both should be reported. This is plausibly implicated in the electricity reversal in Section 2.1.
- **Decision-family generalization.** M5 and PJM are different domains but both instantiate asymmetric quantile decisions. The paper does not establish generalization to storage arbitrage, scheduling, CVaR, or other decision structures.
- Ω **assumptions.** Spectral estimates are unstable for short series, and the source paper recommends a floor of 1000 timesteps. They can also miss regime shifts and exogenous shocks. This is part of why K1 and K3 and the insufficiency test are necessary.
- **Model roster.** Some TSFMs do not expose comparable probabilistic outputs. Wrapping models differently creates a confound, so the final roster must use a consistent quantile interface.
- **Synthetic-to-real gap.** Matched-Ω pairs can prove a controlled failure exists but may not capture every higher-order structure that matters in real data.
- **Cost realism.** M5 and especially PJM require defensible cost ratios. Results should separate normalized decision regret from monetary claims wherever the public cost model is approximate.
- **Schedule.** The submission window is approximately ten weeks. E0 through E4 is already an ambitious workshop paper for a team of four with limited prior research experience. E5 is explicitly deferred for this reason, and E3 is the experiment most likely to consume more time than budgeted.

- **Negative outcomes.** If the scalar boundary is non-monotone or matched-Ω pairs do not flip, the paper narrows to the strongest surviving result rather than adding experiments to manufacture a positive story.

10 Five-Day Feasibility Sprint

Each day has one deliverable and one pass condition. The sprint is not a mini-paper; its job is to falsify expensive assumptions before the full grid is launched.

Day	Work and pass condition
1	Implement and validate Ω against synthetic signals of known structure; execute K1; decide levels versus first differences and document it. *Pass:* pure sine ≈ 0.87, daily plus weekly cycle ≈ 0.79, moderate noise ≈ 0.32, white noise ≈ 0.06.
2	Execute K2 using published GIFT-Eval per-model results. Compute Ω on training splits, join, and check whether the published symmetric-cost relationship reproduces. No forecasting model is run. *Pass:* the published direction and rough magnitude recover. This is the decisive gate.
3	One classical and one TSFM end to end on one development dataset; cache the quantile grid; compute pinball-τ and downstream regret; re-estimate the compute budget from measured throughput. *Pass:* a cached quantile grid exists and full-grid compute is projected within budget.
4	Analytically construct and numerically verify one matched-Ω pair; run structural K3 at $\tau \in \{0.5, 0.91\}$ on a small subset. *Pass:* the Ω match verifies numerically and a boundary can be estimated.
5	Fit the minimal selector on a development subset, freeze it, and smoke-test on a held-out subset. *Pass:* the frozen selector runs end to end on data it has not seen.

Go or no-go memo. Five numbers: Ω on the four synthetic signals against expected; the K2 relationship against the published value; measured throughput extrapolated to the full grid; the verified Ω match for the synthetic pair; and the fraction of oracle value captured by the best scalar threshold at $\tau = 0.91$.

The full experimental grid proceeds only if these gates make the core assumptions credible. Day 2 removes the largest risk in the project for the cost of an afternoon, because the results it needs have already been published by someone else.

11 Deliberately Excluded

The following are outside the main paper unless the core result is complete and a specific reviewer-critical gap remains: rolling Ω, hysteresis, change-point machinery, and switching-frequency analysis; CVaR, repeated-stochastic forecasting, and model-instability studies; additional decision structures such as storage arbitrage; random and MSE selectors and large baseline ladders; equal-weight ensemble competition; conformalized TSFM as a main branch; more than one repair descriptor; and per-model selection claims, since the claim remains family-level with a fixed within-family choice rule.

These are not missing rigor. They are different questions. The paper gets stronger by refusing to answer them at the same time.

12 The Organizing Figure

The page-one figure should be the empirical (Ω, τ) plane: shade the background by family decision advantage, overlay the fitted boundary with uncertainty, mark held-out transfer failures, and overlay matched-Ω counterexamples. A second deployment figure shows the selective-validation frontier, plotting fraction of full validations avoided against excess decision regret. Between these two figures a reader should understand the phenomenon, the failure mode, and the practical payoff without a baseline zoo.

13 What the Final Paper Must Be Able to Say

The paper earns a strong result only if the experiments support an evidence chain rather than a leaderboard win.

1. **Structure.** A stable family-choice boundary exists in the (Ω, τ) plane, and the published symmetric-cost relationship replicates.
2. **Alignment.** Decision-aligned validation at the operational quantile selects differently from generic probabilistic validation, so the cost model carries information CRPS does not.
3. **Transfer.** A frozen minimal selector recovers most of the value of exact pinball-τ validation on datasets it has never seen.
4. **Failure mode.** Matched-second-order processes expose where (Ω, τ) is insufficient, with decision-rank changes that cannot be fixed by a different threshold over the same information.
5. **Deployment value.** The selector knows when to defer, producing a clear validation-compute against decision-regret frontier.

That gives a single story: boundary, then alignment, then transfer, then insufficiency, then safe compute savings. If any one claim fails, the paper narrows around the strongest surviving result rather than adding experiments to hide the failure.

14 Why AAAI-27, and How This Project Gets There

Venue. AI4TS at AAAI-27. Based on the 2026 edition: 4 to 7 pages excluding references, double-blind, an optional appendix of arbitrary length, submissions due early November, and non-archival, so acceptance does not prevent later submission to a larger venue.

Three claims tie this work to that specific room, and all three are verifiable. First, we extend a paper that received an oral slot at this workshop and released its code, so its authors may be in the room or reviewing. Second, we answer a future-work direction those authors named explicitly in print: the paper states that future work should examine probabilistic scores such as CRPS and interval coverage. Our expensive target is pinball-τ, which by the identity in Section 2 is exactly the quantile decomposition of CRPS, so this is that future work carried out and taken one step further. Third, we resolve a contradiction that sits inside the same proceedings, described in Section 2.1: two orals from the same institute on the same day, reaching opposite conclusions on Singapore electricity, neither citing the other.

The audience is already receptive. Of the papers accepted at the 2026 edition, only one scored forecasts by decision quality rather than forecast error. The keynote was titled *Being Small in the Era of Large Models*, and several accepted papers argue that foundation models are oversold. Our central question, when the expensive model family stops being worth its cost, is one this community is already asking.

What acceptance requires of us. Reproduce the Ω relationship in K2 before extending it, since a clean replication is the strongest opening if the original authors review. Fit the page limit, with the (Ω, τ) decision map on page one and the full grid in the appendix. Beat CRPS-based selection, not only error-based selection. Lead the abstract with the decision boundary

and the contradiction it explains, not with the mechanics of the selector. Report money as well as normalized regret for M5 and PJM. Release code, matching the practice of the paper we extend. Budget for one author to attend in person, since there is no virtual presentation option.

Fallbacks. The same organizers run parallel AI4TS editions at IJCAI, SDM, and ICDM. Beyond those, the ICLR 2027 workshop track or a forecasting venue directly. Non-archival status at AAAI keeps all of these available.

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